

59. Fetch API (saare options)

Ye zyaadatar options kam use hote hain. Skip karke bhi `fetch` achhe se use kar sakte ho. Par pata hona achha hai.

Saare options unki default values ke saath:

```
js

fetch(url, {
  method: "GET", // POST, PUT, DELETE, etc.
  headers: { "Content-Type": "text/plain;charset=UTF-8" },
  body: undefined, // string, FormData, Blob, BufferSource,
  referrer: "about:client", // "" ya same-origin URL
  referrerPolicy: "strict-origin-when-cross-origin",
  mode: "cors", // same-origin, no-cors
  credentials: "same-origin", // omit, include
  cache: "default", // no-store, reload, no-cache, force-cache
  redirect: "follow", // manual, error
  integrity: "", // jaise "sha256-abcdef1234567890"
  keepalive: false, // true
  signal: undefined, // AbortController
  window: window // null
});
```

`method`, `headers`, `body` (fetch chapter) aur `signal` (abort chapter) pehle cover ho chuke.

`referrer` aur `referrerPolicy`

`Referer` HTTP header kaise set ho.

- `referrer`: exact `Referer` set karo (sirf current origin ke andar) ya hata do:

```
js

fetch('/page', { referrer: "" }); // koi Referer header nahi
```

- `referrerPolicy`: general rules. 3 tarah ki requests: same origin, dusra origin, HTTPS -> HTTP.

Value	Same origin	Another origin	HTTPS->HTTP
no-referrer	-	-	-
no-referrer-when-downgrade	full	full	-
origin	origin	origin	origin
origin-when-cross-origin	full	origin	origin
same-origin	full	-	-
strict-origin	origin	origin	-
strict-origin-when-cross-origin (default)	full	origin	-
unsafe-url	full	full	full

(full = poora URL, origin = sirf `https://site.com`, `-` = kuch nahi)

Example: admin area ka path bahar ki sites ko nahi dikhana:

```
js
fetch('https://another.com/page', { referrerPolicy: "origin-when-cross-origin"
```

Referrer policy sirf `fetch` ke liye nahi: poore page ke liye `Referrer-Policy` HTTP header, ya link par `<a rel="noreferrer">`.

mode

Galti se hone wali cross-origin requests se bachne ka safeguard:

- `"cors"` (default): cross-origin allowed
- `"same-origin"`: cross-origin **mana**
- `"no-cors"`: sirf safe cross-origin requests

Tab useful jab URL third-party se aaye aur "power off switch" chahiye.

credentials

Cookies aur HTTP-Authorization bhejne ya nahi:

- `"same-origin"` (default): cross-origin par nahi bhejta
- `"include"`: hamesha bhejta hai (cross-origin server se `Access-Control-Allow-Credentials` chahiye)
- `"omit"`: kabhi nahi, same-origin par bhi nahi

cache

Default me standard HTTP caching ((Expires), (Cache-Control), (If-Modified-Since)...).

Options:

- ("default"): standard rules
- ("no-store"): cache poori tarah ignore (agar (If-Modified-Since), (If-None-Match), (If-Unmodified-Since), (If-Match), (If-Range) header ho to ye default ban jaata hai)
- ("reload"): cache se result nahi lega, par response se cache bhar dega (agar headers allow karein)
- ("no-cache"): cached response ho to conditional request, warna normal; cache bharta hai
- ("force-cache"): cache ka response use karo (stale bhi), na ho to normal request
- ("only-if-cached"): cache ka use karo (stale bhi), na ho to error. Sirf (mode: "same-origin") me chalta hai

redirect

- ("follow") (default): redirects (301, 302...) khud follow
- ("error"): redirect par error
- ("manual"): khud sambhalo. Redirect par special response milta hai ((response.type = "opaqueredirect"), status waghera khaali)

integrity

Response ko pehle se pata **checksum** se verify karo. SHA-256, SHA-384, SHA-512 supported.

```
js
```

```
fetch('http://site.com/file', { integrity: 'sha256-abcdef' });
```

Fetch khud SHA-256 nikal ke compare karta hai, mismatch par error.

keepalive

Request page ke **baad bhi** chal sake (page chhodne par bhi). Example: (onunload) me statistics bhejna. Normally page unload par saari related requests abort ho jaati hain, (keepalive) unhe background me poora karta hai:

```
js
```

```
window.onunload = function() {  
  fetch('/analytics', { method: 'POST', body: "statistics", keepalive: true });  
};
```

Limitations:

- Body limit **64KB** (saari `keepalive` requests ka total). Bahut stats ho to beech beech me regularly bhejte raho.
- Document unload ho gaya to **server ka response handle nahi** kar sakte (analytics me usually koi dikkat nahi).